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Week 4&5

TRANSCRIPTOMIC VISUALIZATION AND CLUSTERING ANALYSIS OF THE GSE131907 BREAST CANCER scRNA-seq DATASET

I INTRODUCTION

Breast cancer remains one of the most prevalent malignancies worldwide and represents a major global health burden. According to recent epidemiological studies, breast cancer accounts for a significant proportion of cancer-related morbidity and mortality among women, highlighting the urgent need to better understand its molecular and cellular mechanisms (Sung *et al.*, 2021). Tumor development and progression are influenced not only by malignant epithelial cells but also by the surrounding tumor microenvironment, which consists of immune cells, stromal cells, endothelial cells, and other supporting cellular components.

Traditional bulk RNA sequencing approaches measure average gene expression across heterogeneous cell populations, thereby masking the transcriptional variability present at the single-cell level. Consequently, critical cellular subpopulations and rare cell states that may play key roles in tumor progression, immune response, or therapeutic resistance remain undetected (Luecken & Theis, 2019). In contrast, single-cell RNA sequencing (scRNA-seq) enables high-resolution characterization of gene expression profiles at the individual cell level, allowing researchers to investigate cellular heterogeneity within complex tissues.

Recent advances in single-cell technologies have significantly improved our understanding of tumor ecosystems by revealing diverse cell populations and transcriptional programs within cancer tissues. In breast cancer, scRNA-seq studies have uncovered complex cellular interactions between tumor cells and the surrounding microenvironment, including immune infiltration and stromal remodeling that contribute to tumor progression (Wu *et al.*, 2021). These findings highlight the importance of studying tumors at single-cell resolution to better understand cancer biology.

In this study, a single-cell transcriptomic analysis was performed to investigate cellular heterogeneity in breast cancer samples. The analysis pipeline included quality control, normalization, identification of highly variable genes, dimensionality reduction, clustering, and marker gene identification. Visualization techniques such as UMAP embedding, dot plots, and heatmaps were used to explore gene expression patterns and characterize transcriptionally distinct cell populations. This workflow demonstrates the application of computational bioinformatics approaches to analyze single-cell transcriptomic data and provides insights into the cellular composition of the breast cancer tumor microenvironment.

II METHODS

2.1 Dataset Acquisition

Single-cell RNA sequencing data analyzed in this study were obtained from the publicly available dataset GSE131907 deposited in the Gene Expression Omnibus (GEO) database. The dataset contains transcriptomic profiles from individual cells derived from breast cancer tissues, enabling the investigation of tumor-associated cellular heterogeneity.

2.2 Data Processing and Computational Environment

All analyses were conducted using R within the Seurat framework. Additional packages including ggplot2 and dplyr were used for data visualization and manipulation.

2.3 Quality Control

Quality control was performed to remove low-quality cells and potential technical artifacts prior to downstream analysis. Three key metrics were evaluated:

- number of detected genes per cell (*nFeature_RNA*)
- total RNA counts per cell (*nCount_RNA*)
- percentage of mitochondrial gene expression (*percent.mt*)

Cells with fewer than 200 detected genes, more than 6000 genes, or mitochondrial gene content greater than 10% were filtered out to ensure high-quality data for analysis.

2.4 Data Normalization

Gene expression counts were normalized using a global-scaling normalization method implemented in Seurat. This method normalizes gene expression values for each cell by the total expression, multiplies by a scale factor (10,000), and applies log transformation to stabilize variance across genes.

2.5 Identification of Highly Variable Genes

Highly variable genes (HVGs) were identified to capture biologically meaningful transcriptional variation across cells. These genes exhibit higher dispersion relative to their average expression and are therefore informative for downstream dimensionality reduction and clustering analyses.

2.6 Dimensionality Reduction

Principal Component Analysis (PCA) was performed to reduce the dimensionality of the dataset and identify major sources of variation in gene expression. The optimal number of principal components retained for further analysis was determined using an elbow plot.

2.7 Cell Clustering

Cells were grouped into clusters based on transcriptional similarity using a graph-based clustering approach implemented in Seurat. A shared nearest neighbor (SNN)

graph was constructed based on the selected principal components, and clusters were identified using a modularity optimization algorithm.

2.8 UMAP Visualization

Uniform Manifold Approximation and Projection (UMAP) was applied to visualize the clustered cells in a two-dimensional embedding. This dimensionality reduction method preserves both local and global structure in high-dimensional datasets and allows transcriptionally similar cells to cluster together.

2.9 Marker Gene Identification

Cluster-specific marker genes were identified using differential gene expression analysis implemented in Seurat. Genes significantly upregulated in each cluster were considered candidate markers for defining cluster identity.

III RESULT AND DISCUSSION

3.1 Visualization of Single-Cell Transcriptomic Analysis

The quality control (QC) step is critical in single-cell RNA sequencing analysis to remove low-quality cells and technical artifacts prior to downstream analysis. In Figure 1 QC Violin Plot, three metrics were evaluated: the number of detected genes per cell (*nFeature_RNA*), total transcript counts per cell (*nCount_RNA*), and the percentage of mitochondrial gene expression (*percent.mt*). The distribution of *nFeature_RNA* indicates the complexity of transcriptomic capture in each cell. Most cells in this dataset exhibit approximately 1000–2500 detected genes, suggesting adequate sequencing depth and transcript diversity. Cells with extremely low gene counts are often indicative of dying or damaged cells, while excessively high counts may represent potential doublets or multiplets (Luecken & Theis, 2019).

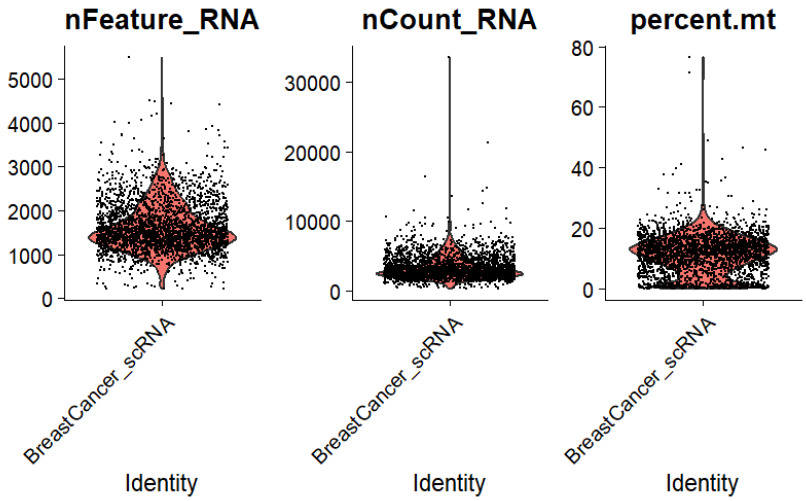

Figure 1 QC violin plot

The *nCount_RNA* distribution reflects the total RNA molecules captured per cell. The majority of cells fall within a moderate range, indicating consistent library preparation and sequencing efficiency across the dataset. Outliers with abnormally high RNA counts may represent doublet events, where two cells are captured within the same droplet during sequencing (Hafemeister & Satija, 2019). The mitochondrial gene percentage (*percent.mt*) serves as an indicator of cell stress or apoptosis. Elevated mitochondrial transcript levels are commonly associated with compromised cell integrity. In this dataset, most cells exhibit mitochondrial proportions below ~20%, suggesting that the majority of cells are of acceptable quality. Cells exceeding typical thresholds (often 10–20%) are generally filtered out during preprocessing to avoid confounding downstream analyses. Overall, the QC assessment shown in Figure QC Violin Plot indicates that the dataset contains a substantial number of high-quality cells suitable for subsequent normalization, clustering, and cell type identification.

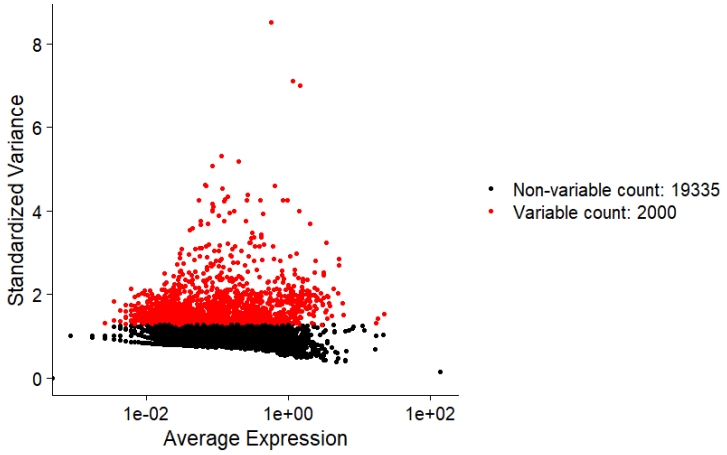

Figure 2 Highly Variable Gene (HVG) Plot

The highly variable gene (HVG) plot illustrates the relationship between average gene expression and the variability of gene expression across all cells in the dataset. In single-cell transcriptomic analysis, identifying highly variable genes is a critical step because only a subset of genes contributes substantially to the biological heterogeneity observed among individual cells. Many genes show relatively stable expression levels across cells and therefore provide limited information for distinguishing cellular populations. In contrast, highly variable genes exhibit significant differences in expression across cells and are often associated with biological processes such as cell differentiation, signaling activity, or cellular responses to environmental conditions (Brennecke *et al.*, 2013). By focusing on these genes, downstream analyses can more effectively capture the transcriptional diversity present within the dataset.

In the HVG plot (Figure 2), the x-axis represents the average expression level of each gene across all cells, while the y-axis represents the standardized variance of gene expression. Each point corresponds to a gene in the dataset. Genes shown in black represent non-variable genes that display relatively low variability in expression across

cells, whereas genes highlighted in red represent highly variable genes that exhibit greater variability than expected based on their mean expression level. According to the figure, approximately 2,000 genes were classified as highly variable, while more than 19,000 genes were considered non-variable. Selecting around two thousand highly variable genes is a commonly adopted strategy in scRNA-seq analysis because it provides a manageable feature set that retains the most informative biological signals while reducing computational complexity.

The distribution pattern observed in the plot reflects a well-known property of RNA sequencing data in which gene expression variance depends on the mean expression level. Genes with very low expression may appear highly variable due to stochastic transcriptional noise and technical sampling effects, while genes with very high expression levels often represent housekeeping genes that maintain stable expression across cells. Therefore, statistical methods are used to standardize variance and identify genes whose variability exceeds what would be expected from their mean expression level alone (Kolodziejczyk *et al.*, 2015). The red points above the main distribution curve represent genes that deviate from this expected trend, indicating genuine biological variability.

The identification of highly variable genes plays a central role in subsequent analytical steps. These genes serve as the primary features used in dimensionality reduction techniques such as PCA, as well as clustering algorithms that aim to identify distinct cell populations within the dataset. By focusing on genes with high variability, the analysis becomes more sensitive to transcriptional differences that reflect underlying biological processes. In the context of this breast cancer dataset, the highly variable genes likely correspond to transcripts involved in tumor cell proliferation, cell–cell communication, and microenvironmental interactions. Consequently, these genes provide the foundation for identifying transcriptionally distinct cellular populations and for detecting marker genes that characterize specific clusters within the tumor microenvironment (Tang *et al.*, 2019).

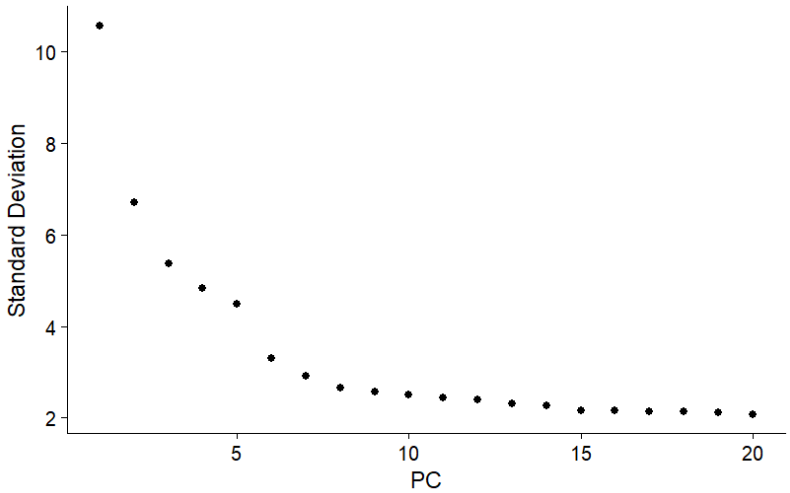

Figure 3 Principal Component Analysis (PCA) Elbow Plot

The principal component analysis (PCA) elbow plot illustrates how much variation in the single-cell RNA sequencing dataset is explained by each principal component. In scRNA-seq analysis, PCA is widely used as a dimensionality reduction technique to summarize complex gene expression profiles into a smaller number of components that capture the most important biological variation among cells. Since scRNA-seq datasets typically measure expression levels of tens of thousands of genes across thousands of cells, the resulting data matrix is highly dimensional and often contains substantial technical noise. PCA addresses this challenge by transforming the original gene expression matrix into orthogonal principal components that successively maximize the variance captured in the dataset (Jolliffe & Cadima, 2016). Through this transformation, correlated gene expression patterns can be summarized into a smaller number of variables, facilitating downstream analyses such as clustering and visualization.

In the elbow plot, the x-axis represents the principal component number, while the y-axis represents the standard deviation associated with each component, which corresponds to the magnitude of variance explained. The figure shows a sharp decrease in standard deviation between the first several components, followed by a gradual flattening of the curve. The first principal component exhibits the largest variance, indicating that it captures the dominant transcriptional pattern within the dataset. Subsequent components contribute progressively smaller amounts of variance, reflecting finer-scale differences between cells. Around the sixth to eighth principal component, the slope of the curve becomes noticeably flatter, forming the characteristic "elbow" shape that indicates diminishing returns in explained variance.

This elbow point is commonly used as a heuristic criterion to determine how many principal components should be retained for downstream analysis. Components before the elbow typically capture biologically meaningful variation related to differences between cell types or cell states, whereas components after the elbow tend to represent minor variation or technical noise. Identifying this transition point is particularly important in single-cell studies because the inclusion of too many components may introduce noise, while using too few may obscure biologically relevant heterogeneity (Andrews & Hemberg, 2018). Based on the pattern observed in this dataset, the elbow appears around PC6–PC8, suggesting that retaining approximately eight to ten principal components would effectively summarize the major transcriptional structure of the dataset. This selection provides a balanced representation of the biological variability present in the breast cancer cells while minimizing the influence of noise in subsequent analyses such as clustering or UMAP projection.

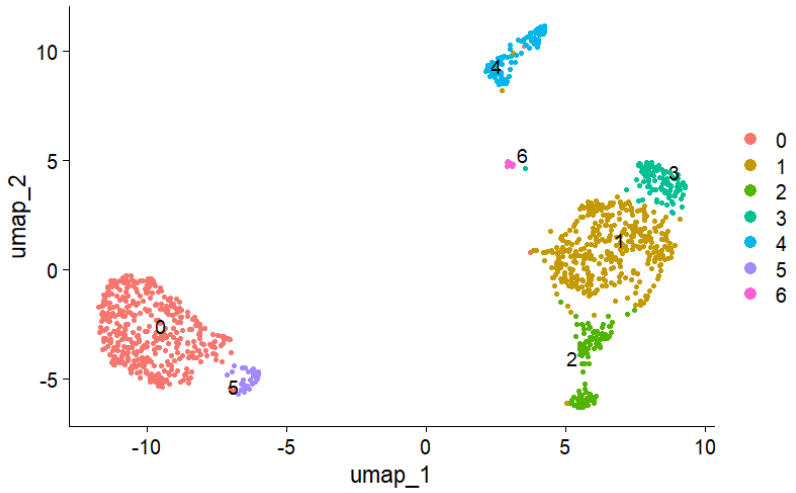

Figure 2 UMAP Visualization of Cell Clusters Based on Highly Variable Genes

Dimensionality reduction using Uniform Manifold Approximation and Projection (UMAP) was performed to visualize transcriptional heterogeneity across individual cells in the dataset. The resulting embedding in Figure 2 demonstrates the spatial separation of cells into seven transcriptionally distinct clusters (clusters 0–6), indicating substantial heterogeneity within the breast cancer cellular population. UMAP is particularly well-suited for single-cell transcriptomic analysis because it preserves both local and global structure of high-dimensional gene expression data, enabling accurate visualization of relationships among cell populations (McInnes *et al.*, 2018). Cells positioned close together in the UMAP space share similar transcriptional profiles, whereas cells located in separate regions represent transcriptionally distinct populations.

In the presented UMAP embedding, cluster 0 forms a large and clearly separated population on the left side of the projection, suggesting a dominant transcriptional program within the dataset. Clusters 1, 2, and 3 appear as intermediate populations, while clusters 4–6 form smaller groups, indicating less abundant cellular subpopulations. Such clustering patterns are commonly observed in tumor tissues, where multiple cell types—including tumor epithelial cells, immune cells, stromal fibroblasts, and endothelial cells—coexist within the tumor microenvironment (Wu *et al.*, 2021). The separation of clusters in Figure 2 indicates that the selected highly variable genes successfully captured biologically meaningful variation among cells. This dimensionality reduction therefore provides a robust framework for subsequent marker gene identification and cluster characterization.

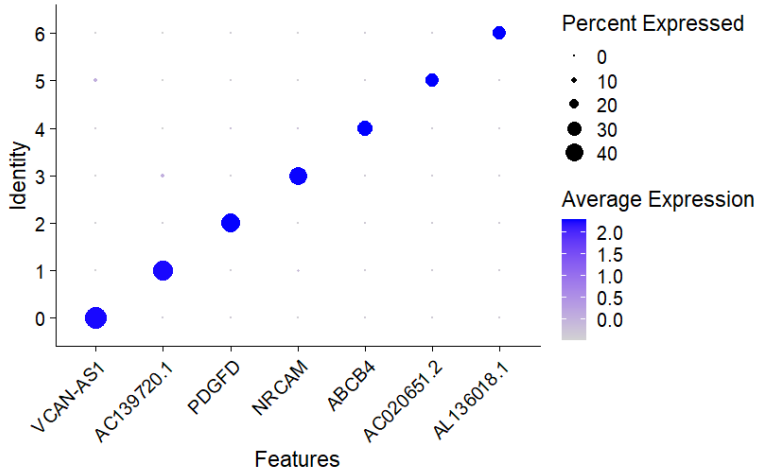

Figure 3 DotPlot of Top Marker Genes

To characterize the transcriptional identity of each cluster, marker gene analysis was performed and the top differentially expressed genes were visualized using a dot plot. Figure 3 displays the expression patterns of selected cluster-specific genes, including *VCAN-AS1, AC139720.1, PDGFD, NRCAM, ABCB4, AC020651.2,* and *AL136018.1*. In this representation, the size of each dot corresponds to the percentage of cells within a cluster expressing the gene, while the color intensity reflects the average expression level. This dual encoding provides a concise summary of both gene prevalence and expression magnitude across clusters (Stuart *et al.*, 2019).

The dot plot reveals that each gene exhibits a cluster-specific expression pattern. For example, *VCAN-AS1* shows high expression predominantly in cluster 0, suggesting that it may serve as a marker gene defining the transcriptional identity of this population. Similarly, *AC139720.1* appears strongly enriched in cluster 1, while *PDGFD* shows elevated expression in cluster 2. Other genes, including *NRCAM*, *AC020651.2*, and *AL136018.1*, display preferential expression in clusters 3, 5, and 6, respectively. Such cluster-specific expression patterns indicate that these genes may play important roles in defining the functional characteristics of each cellular population. Marker genes identified through differential expression analysis are commonly used for cell type annotation by comparing them with known gene signatures from reference datasets (Luecken & Theis, 2019).

Figure 4 Heatmap of Cluster-Specific Top Marker Genes

To further examine gene expression patterns across clusters, a heatmap was generated using the top marker genes identified from differential expression analysis. Figure 4 visualizes the normalized expression levels of selected genes across individual cells grouped by cluster identity. In the heatmap, rows correspond to genes while columns represent individual cells. The color gradient indicates relative expression levels, with brighter colors representing higher gene expression. Distinct expression blocks can be observed across clusters, indicating that specific genes are preferentially expressed within particular cell populations. For instance, *VCAN-AS1* exhibits strong expression primarily in cluster 0, forming a distinct block of high expression in that region of the heatmap. Similarly, *AC139720.1* appears highly expressed in cluster 1, while *PDGFD* demonstrates increased expression in cluster 2. These patterns confirm the cluster-specific gene expression observed in the dot plot and further validate the marker gene selection.

Heatmaps are widely used in single-cell transcriptomic analysis because they provide a global overview of transcriptional programs across clusters and allow rapid identification of genes that define specific cellular populations (Luecken & Theis, 2019). The distinct expression signatures observed in Figure 4 suggest that each cluster possesses a unique transcriptional identity. Collectively, the results from the dot plot and heatmap analyses demonstrate that the clustering approach successfully identified transcriptionally distinct cell populations within the dataset. These marker genes may serve as candidate biomarkers for further investigation of cellular heterogeneity within the breast cancer microenvironment.

3.2 Biological Interpretation of Identified Marker Genes

Beyond identifying transcriptionally distinct clusters, interpreting the biological roles of the top marker genes provides insight into the functional heterogeneity of the cellular populations observed in the dataset. Several genes

detected in the analysis are known to be associated with tumor biology, cellular signaling, and microenvironment interactions.

One of the identified marker genes is PDGFD (Platelet Derived Growth Factor D), which plays an important role in cell proliferation, angiogenesis, and stromal cell recruitment. PDGFD signaling has been implicated in tumor progression and remodeling of the tumor microenvironment by promoting fibroblast activation and extracellular matrix production. Increased expression of PDGFD in specific clusters may therefore indicate the presence of stromal or fibroblast-like cell populations that contribute to tumor growth and tissue remodeling.

Another gene detected in the analysis is NRCAM (Neuronal Cell Adhesion Molecule), which is involved in cell-cell adhesion and signaling processes. Although originally characterized in neural tissues, NRCAM has been increasingly reported in cancer studies where it contributes to tumor invasion, metastasis, and altered cell adhesion dynamics. Its presence in certain clusters suggests that these cell populations may possess enhanced migratory or invasive properties.

The gene ABCB4, a member of the ATP-binding cassette transporter family, is associated with membrane transport processes. ABC transporters are known to regulate the movement of lipids, metabolites, and xenobiotics across cellular membranes. In cancer biology, several ABC transporters are linked to multidrug resistance mechanisms, where tumor cells actively export chemotherapeutic agents to survive treatment. Elevated expression of ABCB4 in specific clusters could therefore reflect metabolic specialization or adaptive survival mechanisms within certain cell populations.

Additionally, non-coding RNA genes such as VCAN-AS1 and AC139720.1 were identified as cluster-specific markers. Long non-coding RNAs (lncRNAs) have emerged as important regulators of gene expression, chromatin organization, and cellular signaling pathways. Increasing evidence suggests that lncRNAs play significant roles in cancer progression by modulating transcriptional programs and interacting with protein complexes involved in tumorigenesis.

Taken together, the expression patterns of these marker genes suggest that the clusters identified in the dataset likely represent functionally distinct cellular populations within the breast cancer microenvironment. These populations may include tumor epithelial cells, stromal fibroblasts, or other supportive cell types that collectively contribute to tumor heterogeneity. Understanding the biological significance of these marker genes is therefore crucial for interpreting the cellular composition and functional diversity captured by single-cell transcriptomic analysis.

IV Conclusion

Single-cell transcriptomic analysis of the breast cancer dataset GSE131907 reveals substantial cellular heterogeneity within the tumor microenvironment. The bioinformatics workflow involving quality control, normalization, identification of highly variable genes, dimensionality reduction, clustering, and marker gene detection successfully identifies transcriptionally distinct cell populations. Dimensionality reduction using PCA and visualization with UMAP separates the cells into seven

distinct clusters, indicating diverse transcriptional profiles across the dataset. Marker gene analysis highlights several cluster-specific genes, including PDGFD, NRCAM, and ABCB4, as well as non-coding RNAs such as VCAN-AS1 and AC139720.1. These genes are associated with biological processes including cell proliferation, adhesion, transport mechanisms, and transcriptional regulation, which are relevant to tumor progression and microenvironment interactions. Single-cell RNA sequencing combined with computational analysis using Seurat provides a powerful approach for identifying transcriptional diversity within breast cancer tissues and offers valuable insights into the cellular composition of the tumor microenvironment.

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